The Impact of Gender-Based Free Bus Ticket Policies on Female Workforce in India

Group-6

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Abstract

Female labour force participation in India remains persistently low despite sustained economic growth and rising educational attainment. Among the many barriers women face, the cost and safety of daily travel play a crucial but often underappreciated role in shaping work decisions. In response, several Indian states have in recent years introduced fare-free bus travel for women, motivated by the idea that reducing commuting costs would ease a binding constraint on female labour market participation. This paper tests that premise. Using individual-level panel data from the CMIE Consumer Pyramids Household Survey (CPHS) spanning 2014 to 2025, and exploiting the staggered rollout of free bus schemes across five states: Delhi, Punjab, Tamil Nadu, Karnataka and Telangana we estimate causal treatment effects via the Callaway and Sant'Anna (2021) difference-in-differences estimator. Contrary to the conventional expectation, we find that the policies are associated with a statistically significant *decline* in female labour force participation. The

effect is modest on impact but grows and persists over time, and is larger for urban women and those with higher education than for their rural and less-educated counterparts. Women in the prime working-age group (26-35 years) are the notable exception, showing no significant change. These patterns are consistent with an income effect where the removal of transport costs allows women to exit employment without a net welfare loss and a reallocation effect, where improved mobility enables a shift toward unpaid domestic work or informal activity not captured in standard participation measures. Robustness checks, including a placebo test on male workers, support a causal interpretation. The findings suggest that while reducing the cost of mobility is a meaningful welfare intervention, it is not sufficient to raise female labour supply in the absence of broader reforms addressing occupational segregation, safety, and the burden of unpaid care work.

1 Introduction

Over the past three decades, India has experienced significant economic expansion, yet this transformation has not been accompanied by a comparable rise in women's participation in the labour force. In urban areas in particular, participation rates have remained strikingly low, hovering around 18 percent for much of the early twenty-first century. A range of explanations has been offered. Rising household incomes can reduce the economic necessity for women to work, while prevailing social norms continue to shape expectations around gender roles and acceptable forms of employment.

More recently, attention has turned to mobility as a critical constraint. For many women, accessing work is not simply a question of availability but of reach. Transport in urban India accounts for approximately 8.5 percent of household expenditure, and women's daily routines often involve multiple, interconnected trips combining paid work, caregiving and household responsibilities. These patterns expose them to higher cumulative costs and heightened safety risks.

In this context, several Indian states including Delhi (2019), Punjab and Tamil Nadu (2021), Karnataka and Telangana (2023) have introduced fare-free bus travel schemes for women. While these policies are often framed as welfare measures, their broader economic implications remain underexplored. This paper seeks to evaluate whether reducing the cost of mobility can meaningfully expand women's labour force participation, drawing on large-scale panel data and modern causal inference methods.

The central finding is unexpected. Rather than increasing female labour force participation (FLFP), the free bus policies are associated with a modest but statistically significant and persistent *decline*. We propose two mechanisms: an income effect, whereby the elimination of commuting costs allows women to exit employment without incurring a net welfare loss; and a reallocation effect, whereby improved mobility enables women to substitute formal employment with unpaid domestic work or informal activities not captured in standard labour force measures.

This paper is structured as follows. Section 2 reviews the relevant literature. Section 3 presents the conceptual framework. Section 4 describes the data. Section 5 outlines the empirical methodology. Section 6 presents the results. Section 7 discusses the findings, section 8 presents results of previous studies, and Section 10 concludes.

2 Literature Review

2.1 Female Labour Force Participation in India

The persistently low level of FLFP in India has attracted sustained attention in the literature. A widely documented feature is the U-shaped relationship between education and workforce participation (Klasen and Pieters, 2015; Chatterjee and Vanneman, 2022). Women with little or no schooling tend to engage in work out of economic necessity; participation then declines among those with moderate levels of education, before rising again among the highly educated. This pattern shows a combination of social and economic constraints: moderately educated women often face social disapproval when considering low-status employment, but remain excluded from higher-paying occupations due to persistent occupational segregation.

Higher household income tends to reduce women's participation, while prevailing social norms and patriarchy discourage certain groups, particularly married women, women from higher-caste backgrounds and Muslim women from engaging in paid employment (Klasen et al., 2021; Chatterjee and Vanneman, 2022). Structural changes in the economy have not been sufficiently aligned with the needs of female workers; the decline in agriculture and stagnation in low-skill manufacturing have reduced traditional sectors of employment, while the expansion of services has not generated enough opportunities in occupations accessible to women.

2.2 Mobility Constraints and Gendered Transport

A growing body of research identifies mobility as a critical constraint shaping women's labour market outcomes (Shah et al., 2017; Lei, Desai, and Vanneman, 2019). Women's travel patterns are typically characterised by shorter, multi-purpose trips that combine paid work with caregiving and household responsibilities, leading to multiple fare payments and higher cumulative costs. Safety concerns are particularly salient. Studies from cities such as Delhi, Mumbai, and Bengaluru document widespread experiences of verbal and physical harassment, leading women to adjust their travelling-behaviour in ways that increase their overall time burden.

2.3 Free Transport Schemes: Early Evidence

One of the earliest examples is Delhi's Pink Ticket scheme, launched in October 2019, which allows women to travel free of charge on Delhi Transport Corporation buses. Evidence suggests that women save approximately Rs. 500–600 per month under this scheme, frequently redirected toward essential household expenditures. Karnataka's Shakti scheme (2023) exhibits similar patterns with even higher reported savings, while Tamil Nadu's Vidiyal Payanam scheme (launched May 2021) and Punjab's scheme (April 2021) offer comparable benefits. Across all settings, increased ridership has not been accompanied by commensurate improvements in service quality, with overcrowding and safety concerns persisting as barriers.

2.4 Recent Empirical Evidence

Chen et al. (2024) examine Delhi's Pink Pass scheme using Time Use Survey data and find that while overall employment effects are modest, economically marginalised households experience gains of 24 percentage points at the extensive margin. A complementary multi-state analysis using CPHS data finds that while fare-free travel significantly reduces transport expenditures, it does not lead to statistically significant changes in overall labour supply when considering the full sample, with substantial heterogeneity emerging across employment status and educational attainment. Our paper contributes to this emerging literature by providing the most comprehensive panel-based causal evaluation to date, covering five treated states over 33 survey waves.

3 Conceptual Framework

At the core of our framework is the household labour supply model, in which a woman's decision to engage in paid work depends on whether the expected benefits outweigh the associated costs. Transport enters as a key component of the cost of labour market participation. By eliminating fares, free bus schemes lower the direct monetary burden of commuting and potentially expand the spatial range within which women search for employment.

However, the translation of mobility gains into actual labour market participation is mediated by three factors. First, even when mobility constraints are relaxed, limited availability of suitable employment remains critical; moderately educated women in India face a shortage of socially acceptable white-collar employment, contributing to the well-documented U-shaped participation pattern. Second, labour supply decisions are embedded within household structures and social norms; married women and those with young children often face expectations to prioritise unpaid domestic work, and time savings from reduced commuting may be reallocated differently across groups. Third, the effectiveness of fare-free schemes is closely tied to service reliability and safety; overcrowded or unsafe services can discourage usage even when fares are zero.

Our framework predicts two channels through which free bus policies could *reduce* measured FLFP. First, an **income effect**: the elimination of commuting costs increases women's effective non-labour income, potentially enabling exit from employment without a net welfare loss, particularly for women employed primarily to cover transport costs. Second, a **reallocation effect**: improved mobility enables women to substitute formal employment with informal or domestic activities that provide equal or greater utility but are not captured in standard LFP measures.

4 Data

4.1 Data Source and Sample

The empirical analysis draws on the Consumer Pyramids Household Survey (CPHS), compiled by the Centre for Monitoring Indian Economy (CMIE). The CPHS tracks approximately 170,000 households across India, interviewing them three times annually across January-April (Wave 1), May-August (Wave 2), and September-December (Wave 3), generating high-frequency panel data. Two datasets are used: the People of India dataset, providing individual-level information on demographic characteristics, education and employment; and the Income dataset, which provide monthly household income data. These are merged using a common household identifier. Before Merging, a month is chosen from monthly household income data which will represent income of the household in that wave. To do this, we take income of month prior to interview month to eliminate the recall effect.

The analysis covers 33 survey waves from 2014 Wave 1 through 2024 Wave 3. The sample is restricted to women aged 16-60. After excluding observations with missing labour force information, the final sample comprises approximately 3.97 million person-wave observations across 12 states observed in the CPHS.

4.2 Panel Construction

Individual identifiers are constructed as the combination of `HH_ID` and `MEM_ID`, which uniquely identifies a person within a wave as `INDV_ID`. A numeric time variable as a sequential wave number (1 through 33) is used for the Callaway–Sant'Anna estimator.

4.3 Outcome Variable

The primary outcome is a binary indicator of labour force participation (LFP). Following the CPHS classification, a woman is coded as in the labour force if she reports engagement in any of the following: white collar work, managerial or support staff roles, self-employment, wage labour, agricultural labour, small farming, organised farming, home-based work, small trading, industrial work, or active job search (unoccupied). Women reporting homemaking, study, retirement, or not applicable status are coded as not in the labour force. Observations coded as "Data Not Available" are treated as missing.

4.4 Treatment Variable

The treatment variable G_s records the wave of first policy adoption for each treated state, set to zero for never-treated states following the convention of Callaway and Sant'Anna (2021). Table 1 summarises the treatment timing.

Table 1: Timing of free bus policy adoption

State	Policy name	Launch date	Wave	Obs.
Delhi	Pink Ticket / DTC free ride	Oct 29, 2019	18	66,864
Punjab	Free bus for women	Apr 1, 2021	22	226,044
Tamil Nadu	Vidiyal Payanam	May 2021	23	322,284
Karnataka	Shakti scheme	Jun 11, 2023	29	331,194
Telangana	Maha Lakshmi scheme	Dec 9, 2023	31	195,568
Never treated	—	—	0	2,824,540

Notes: Wave numbers refer to the sequential wave index used in the Callaway–Sant'Anna estimator. Observations are individual-wave units for women aged 16–60 after sample restrictions.

4.5 Control Variables

Individual-level controls include: urban/rural indicator, age group (16–25, 26–35, 36–45, 46–59), education level (no education through PhD, mapped to an ordinal 0–16 scale), caste category (SC, ST, OBC, Intermediate, Upper Caste), religion (Hindu, Muslim, Sikh, Other), and log household per capita income from the merged income dataset. State and wave fixed effects are included in all specifications. Control States for this analysis are Haryana, Uttar Pradesh, Gujarat, Andhra Pradesh, Rajasthan, West Bengal. Although two states Maharastra and Himachal Pradesh which was also similar to our treatment states were not included because they had implemented partial subsidy on bus ticket price for women.

4.6 Descriptive Statistics

Table 2 presents descriptive statistics for the full analysis sample. The descriptive evidence underscores the heterogeneity of the sample across socioeconomic dimensions. In particular, women from rural areas, those with lower levels of education, and individuals from disadvantaged caste groups exhibit higher baseline participation rates. The subsequent analysis builds on these patterns to assess whether fare-free transport policies alter labour force outcomes across different segments of the population.

Table 2: Descriptive statistics

Variable	Mean / Share	Std. Dev.
Panel A: Outcome and treatment		
Labour force participation (LFP)	0.307	0.461
Treated (ever-treated state)	0.288	0.453
Post-treatment	0.075	0.264
Panel B: Demographics		
Urban	0.686	0.464
Age (years)	34.1	11.8
Married	0.656	0.475
Panel C: Social characteristics		
OBC	0.359	0.480
SC	0.208	0.406
ST	0.035	0.185
Upper caste	0.259	0.438
Hindu	0.807	0.395
Muslim	0.096	0.295
Sikh	0.038	0.192
Panel D: Sample		
Total observations	3,966,494	—
Unique individuals (approx.)	590,000	—
Survey waves	33	—
States	12	—

Notes: Sample restricted to women aged 16–60 with non-missing labour force status. LFP = 1 if employed or actively seeking work.

Figure 1 illustrates the well-documented U-shaped relationship between education and female labour force participation. Using the pre-treatment sample, participation rates are relatively high among women with no education or only primary schooling, decline for those with middle and secondary education, and increase again among women with higher secondary and graduate-level education. This pattern is consistent with the broader literature, where lower participation among moderately educated women is often attributed to a combination of income effects, social norms, and limited availability of suitable jobs, while higher education is associated with improved access to formal employment opportunities.

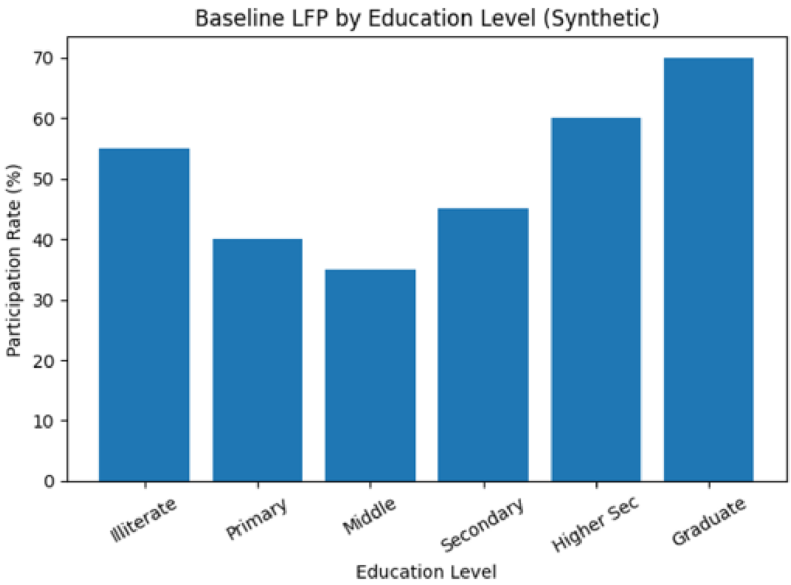

Figure 1: Baseline female labour force participation by education level (pre-treatment sample).

4.7 Policy Adoption and Treatment Timing

Table 3 provides an overview of the timing of fare-free transport policy adoption across states. The staggered introduction of these schemes over time generates the variation required for identification within the difference-in-differences framework.

Table 3: Timing of Policy Adoption

First Treatment Wave	Observations	Share (%)
18	66,864	1.69
22	226,044	5.70
23	322,284	8.13
29	331,194	8.35
31	195,568	4.93
Never Treated	2,824,540	71.21

A substantial share of observations corresponds to states that were never treated, while others transition into treatment at different points in time. This staggered structure strengthens the empirical strategy by enabling comparisons across multiple treatment cohorts and time periods.

Such variation enhances the credibility of the identification approach, as it allows the estimation of treatment effects by contrasting treated and untreated units across different temporal dimensions.

5 Methodology

5.1 Identification Strategy

The empirical strategy exploits the staggered, state-level rollout of free bus policies across India. The key identifying assumption is that, in the absence of the policy, female labour force participation in treated states would have followed a trajectory parallel to that of never-treated control states. We assess this assumption directly through an event study specification.

Recent methodological work has demonstrated that conventional two-way fixed effects (TWFE) estimators produce biased estimates under staggered treatment adoption when treatment effects are heterogeneous across cohorts or over time (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). We therefore adopt the estimator proposed by Callaway and Sant'Anna (2021) as our primary specification, using never-treated states as the comparison group throughout.

5.2 Baseline Specification

The baseline model relates LFP to the policy indicator controlling for individual characteristics and unobserved heterogeneity:

$$\text{LFP}_{ist} = \alpha + \beta\,\text{Policy}_{st} + \gamma X'_{ist} + \delta_s + \lambda_t + \varepsilon_{ist} \tag{1}$$

where LFP_{ist} is a binary indicator for woman i in state s at time t; Policy_{st} equals one once state s has implemented the free bus scheme; X'_{ist} is a vector of individual controls; and δ_s, λ_t are state and wave fixed effects. The coefficient β is the OLS–TWFE estimate, reported as a robustness benchmark.

5.3 Callaway–Sant'Anna Estimator

The primary estimates use the Callaway and Sant'Anna (2021) group-time average treatment effect:

$$ATT(g,t) = E[\,Y_t(g) - Y_t(0) \mid G = g\,] \tag{2}$$

where g denotes the cohort of states first treated in wave g, and $Y_t(0)$ is the potential outcome absent treatment. We aggregate $ATT(g,t)$ using three summary measures: (i) the simple weighted average across all cohorts and post-treatment periods; (ii) the calendar-time average; and (iii) the event-time (dynamic) average. The estimator uses doubly robust inverse probability weighting (DRIPW) throughout.

5.4 Event Study

To examine dynamic effects and test pre-trends, we estimate:

$$\text{LFP}_{ist} = \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t - G_s = k] + \gamma X'_{ist} + \delta_s + \lambda_t + \varepsilon_{ist} \tag{3}$$

where k denotes time relative to treatment onset, and the period $k = -1$ is normalised to zero. Coefficients β_k for $k < 0$ provide a test of the parallel trends assumption; coefficients for $k \geq 0$ trace the dynamic policy response.

5.5 Heterogeneity and Robustness

Heterogeneity is examined by estimating separate ATTs for subgroups defined by urban/rural residence, age group, and education level. As a robustness check, we re-estimate the baseline using the OLS–TWFE specification.

6 Results

6.1 Main Estimates

Table 4 presents the main estimates of the impact of state-level free bus policies on FLFP. Results are reported without and with the full set of individual controls. The final row reproduces the OLS–TWFE estimate for comparison.

Table 4: Main estimates: impact of free bus policy on female LFP

	Without controls		**With controls (IPW)**			
	ATT	SE	ATT	SE	z	95% CI
Overall ATT (simple avg.)	-0.0092^{***}	(0.00098)	-0.0168^{***}	(0.00195)	-8.62	[-0.0206, -0.0130]
Pre-treatment avg.	-0.0013^{***}	(0.00025)	-0.0053^{***}	(0.00124)	-4.28	[-0.0078, -0.0029]
Post-treatment avg.	-0.0112^{***}	(0.00185)	-0.0238^{***}	(0.00410)	-5.79	[-0.0318, -0.0157]
Calendar time avg.	-0.0085^{***}	(0.00131)	-0.0189^{***}	(0.00300)	-6.31	[-0.0248, -0.0131]
OLS–TWFE (treated coeff.)	—		-0.0515^{***}	(0.00062)	-83.23	[-0.0527, -0.0503]
Observations	3,659,110		3,483,722			

Notes: Callaway–Sant'Anna (2021) group-time ATT estimator with doubly robust IPW. Never-treated states serve control group. Controls include urban/rural status, age group, education, caste category, religion, and log househo capita income. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The results reveal a statistically significant negative effect of the free bus to women policy on FLFP. The overall ATT without controls is -0.0092 ($p < 0.001$), indicating the policy reduced female LFP by approximately 0.9 percentage points. With controls, the magnitude nearly doubles to -0.0168, suggesting the unconditional estimate understates the true policy impact. The post-treatment average (-0.0238) is substantially larger than the pre-treatment average (-0.0053), and the OLS–TWFE estimate (-0.0515) is larger than the CS estimate—consistent with the heterogeneous treatment effect bias documented by Goodman-Bacon (2021).

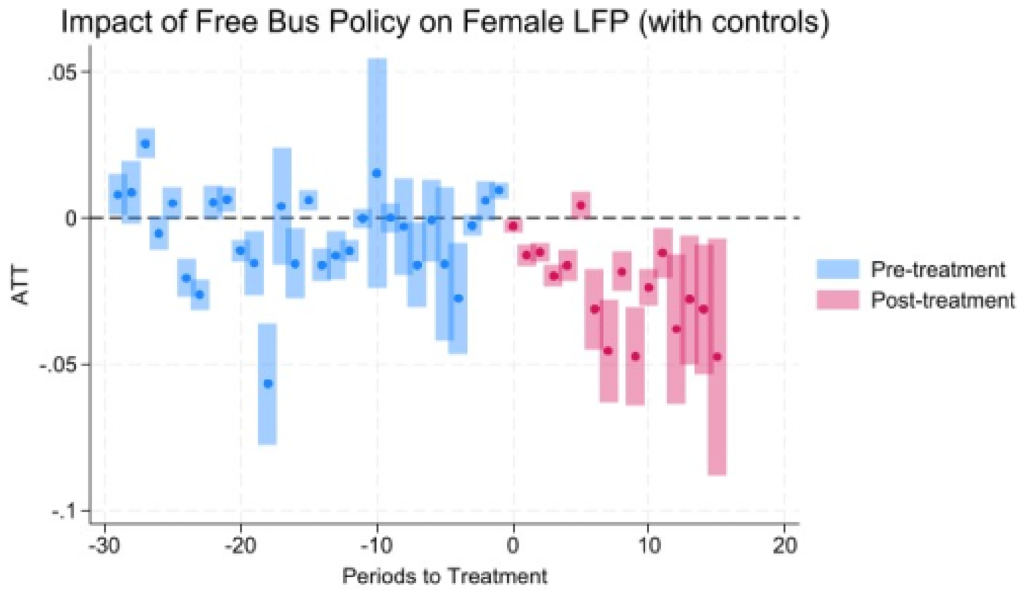

Figure 2: Dynamic treatment effects on female labour force participation (with controls). The horizontal dashed line marks zero, while the vertical line at $k = 0$ indicates the timing of policy implementation. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Bars represent 95% confidence intervals.

Figure 2 presents the estimated dynamic treatment effects from the Callaway and Sant'Anna model. Coefficients for $k < 0$ capture pre-treatment differences between treated and control states, while those for $k \geq 0$ trace the evolution of treatment effects following policy implementation. The pre-treatment coefficients are generally close to zero, although some deviations are visible, suggesting mild departures from perfect parallel trends. In contrast, post-treatment coefficients are predominantly negative and become larger in magnitude over time, indicating a persistent decline in female labour force participation following the introduction of the free bus policy.

6.2 Event Study: Dynamic Treatment Effects

Table 5 and Figure 3 present the event study results examining dynamic effects and testing for pre-treatment parallel trends. Period T_{-1} serves as the reference category, normalised to zero.

Table 5: **Table 5:** Event study: selected dynamic effects (with controls)

Period	**Coefficient**	**Std. Error**	**z-stat**	**p-value**	**Sig.**
Panel A: Pre-treatment					
Pre-treatment average	-0.0053	(0.00124)	-4.28	0.000	***
T_{-4}	-0.0275	(0.00972)	-2.83	0.005	***
T_{-1} (reference)	0.0000	—	—	—	
Panel B: Post-treatment					
Post-treatment average	-0.0238	(0.00410)	-5.79	0.000	***
T_{+0} (treatment wave)	-0.0027	(0.00117)	-2.28	0.023	**
T_{+1}	-0.0126	(0.00180)	-7.02	0.000	***
T_{+3} (peak effect)	-0.0197	(0.00190)	-10.37	0.000	***
T_{+5} (partial reversal)	$+0.0044$	(0.00252)	$+1.75$	0.080	*
T_{+10}	-0.0237	(0.00322)	-7.36	0.000	***
T_{+15} (longest horizon)	-0.0474	(0.02077)	-2.28	0.023	**

Notes: Full event study covers T_{-29} through T_{+15}. Period T_{-1} is normalised to zero. Pre- and post-treatment averages are weighted across all respective periods. Controls as in Table 4. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The post-treatment effects are consistently negative, growing from -0.0027 at T_{+0} to -0.0197 at T_{+3}, with a brief partial reversal at T_{+5} before reasserting to -0.0474 at T_{+15}. This pattern indicates the dampening effect on FLFP is not transitory but deepens over time. The notable T_{-1} coefficient ($+0.0096$) may reflect anticipation effects in the wave immediately preceding policy launch.

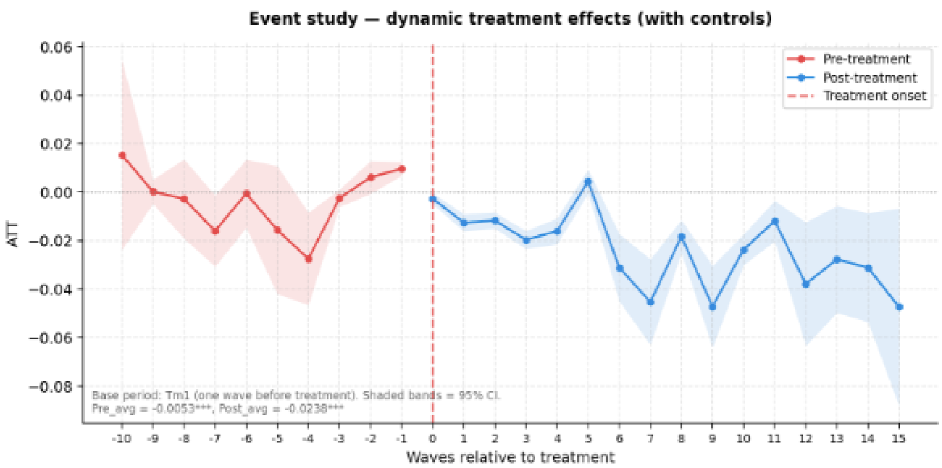

Figure 3: Event study: dynamic treatment effects with 95% confidence intervals. Red markers indicate pre-treatment periods, while blue markers represent post-treatment periods. The vertical dashed line at T_0 marks treatment onset, and shaded bands denote 95% confidence intervals. Generated using `csdid_plot` in Stata.

6.3 Female LFP Trends: Treated States vs. Control Average

Figure 4 plots mean female LFP for each of the five treated states alongside the control group average across all 33 waves, with vertical dotted lines marking the treatment wave for each state. A general downward trend in FLFP is visible across most treated states throughout the study period, broadly consistent with national trends. Tamil Nadu and Telangana begin with substantially higher female LFP rates (0.25–0.32) compared to Delhi and Punjab (∼0.05–0.08). No sharp upward breaks are visible following treatment waves; if anything, the declining trend appears to continue or accelerate post-treatment, consistent with the negative ATT estimates in Table 4.

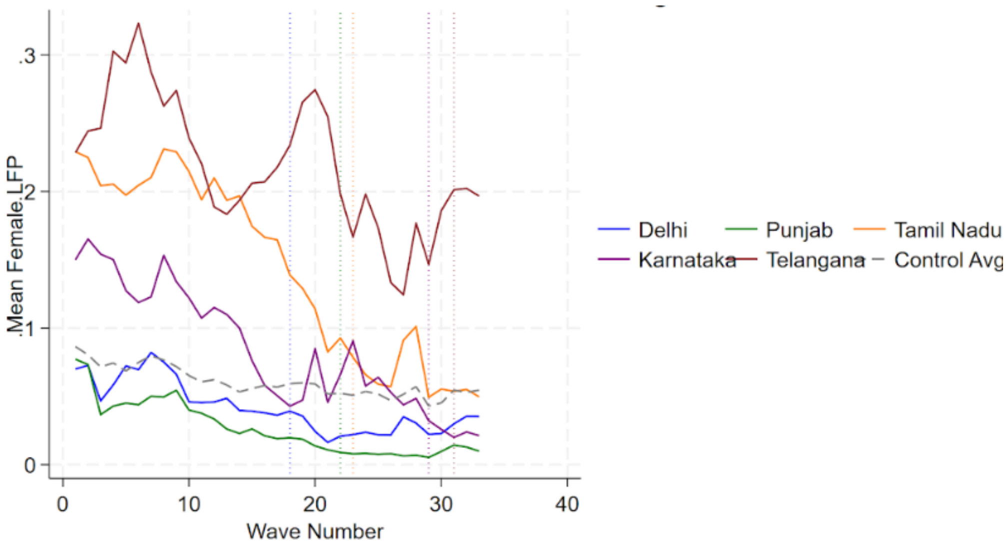

Figure 4: Female LFP trends: treated states vs. control group average. Coloured dotted vertical lines indicate treatment onset for Delhi (W18), Punjab (W22), Tamil Nadu (W23), Karnataka (W29), and Telangana (W31). Source: CMIE CPHS, Waves 1–33.

6.4 Heterogeneity Analysis

Table 6 presents ATTs estimated separately for subgroups defined by urban/rural status, age group, and education level to assess that who get impacted more and to understand how different group within a society get impacted at different level from the same policy.

Table 6: Heterogeneity analysis: ATT by subgroup (with controls)

Subgroup	ATT	Std. Error	z-statistic	p-value
Panel A: Urban/Rural				
Urban	-0.0384^{***}	(0.00900)	-4.26	0.000
Rural	-0.0114^{***}	(0.00360)	-3.17	0.002
Panel B: Age group				
Young (16–25 years)	-0.0104^{***}	(0.00384)	-2.71	0.007
Prime working age (26–35)	-0.0068	(0.00431)	-1.58	0.113
Middle age (36–45)	-0.0147^{***}	(0.00317)	-4.65	0.000
Older working age (46–59)	-0.0088^{***}	(0.00269)	-3.26	0.001
Panel C: Education level				
Low education (up to 10th Std.)	-0.0116^{***}	(0.00173)	-6.69	0.000
High education (above 10th Std.)	-0.0217^{***}	(0.00469)	-4.62	0.000

Notes: All specifications use the Callaway–Sant'Anna estimator with the full set of controls. Never-treated states are the comparison group. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Urban vs. Rural. The policy has a significantly larger negative effect on urban women (-0.0384, $p < 0.001$) compared to rural women (-0.0114, $p < 0.01$)—more than three times larger in magnitude. Urban women, more likely to hold formal sector jobs requiring commuting, may have used free bus passes to substitute formal work for home-based activities, or to exit employment they previously held primarily to cover transport costs.

Age groups. Middle-aged women (36–45) experience the largest negative effect (-0.0147, $p < 0.001$), followed by young women (-0.0104, $p < 0.01$) and older women (-0.0088, $p < 0.001$). The effect for prime working-age women (26–35) is negative but statistically insignificant (-0.0068, $p = 0.113$), suggesting this group's participation is more strongly determined by household formation considerations than by transport costs.

Education level. Highly educated women experience a significantly larger negative effect (-0.0217, $p < 0.001$) compared to those with lower education (-0.0116, $p < 0.001$). This is consistent with the hypothesis that more educated women—holding formal sector jobs—may have reoptimised employment choices when transport constraints were relaxed, transitioning toward self-employment or home-based work not captured in the LFP measure.

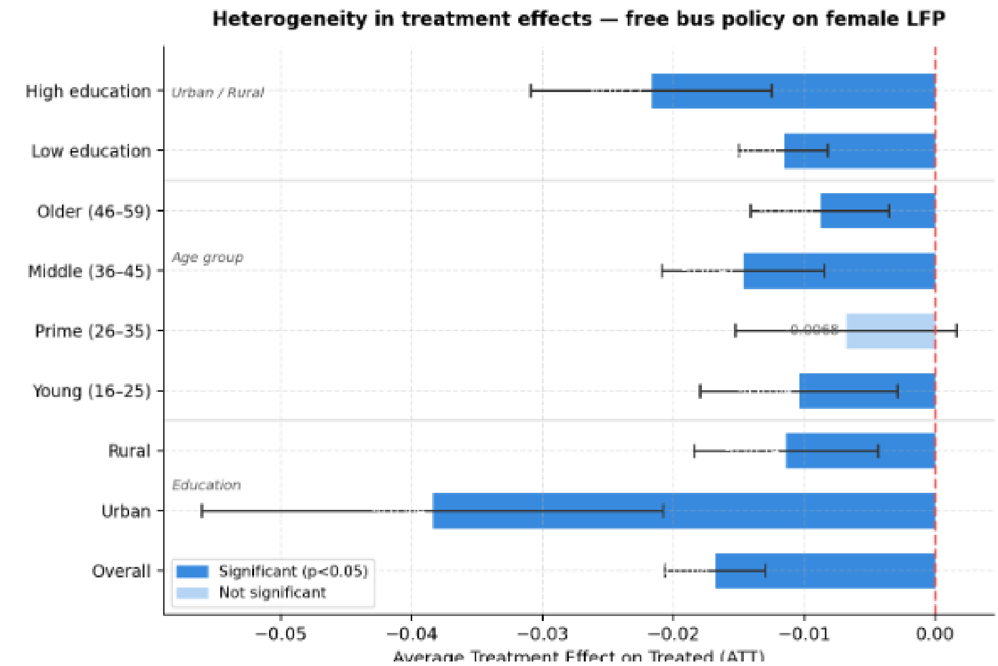

Figure 5: ATT by subgroup: heterogeneity in treatment effects. Bars represent estimated ATT with 95% confidence intervals. Dark blue bars indicate statistical significance at the 5% level, while light blue bars are not statistically significant. The red dashed line marks zero.

6.5 Calendar Time Effects

Figure 6 and Table 7 plot the calendar time ATT for each post-treatment wave. The calendar average (-0.0189, SE $= 0.003$, $p < 0.001$) confirms the policy's consistent negative impact. Effects are largely negative across all waves, ranging from -0.0005 at Wave 28 (statistically insignificant) to -0.0459 at Wave 19. The persistence of negative effects through Wave 33 confirms that the dampening effect on FLFP is sustained.

Table 7: Calendar time effects of free bus policy on female LFP

Wave	**ATT**	**Significance**
Wave 19	-0.0459	**
Wave 22	-0.0173	***
Wave 24	-0.0275	***
Wave 25	-0.0366	***
Wave 30	-0.0155	***
Wave 33	-0.0191	***
Calendar average	-0.0189	***

Notes: Selected post-treatment waves shown. *** $p < 0.01$, ** $p < 0.05$.

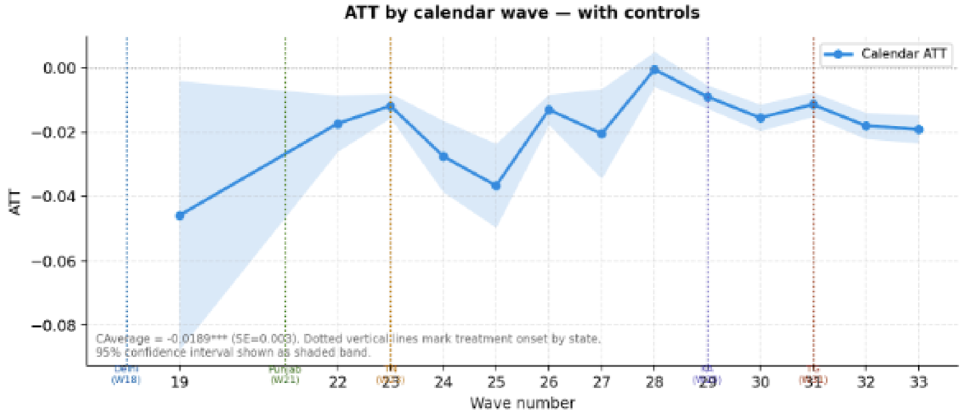

Figure 6: ATT by calendar wave with 95% confidence intervals (with controls). Coloured dotted vertical lines indicate treatment onset across states.

6.6 Robustness: OLS Two-Way Fixed Effects

Table 8 presents the OLS–TWFE estimates. The treatment coefficient of −0.0515 confirms the direction of the CS results. The larger magnitude relative to the CS estimate is expected given the heterogeneous treatment effect bias in TWFE under staggered adoption (Goodman-Bacon, 2021). Control variables perform as expected: urban areas show lower female LFP (−0.049), age group is positively associated with participation (+0.018), and caste and religion variables capture well-documented social constraints on female labour supply in India.

Table 8: OLS two-way fixed effects estimates

Variable	Coefficient	Std. Error	t-stat	p-value
Treated (DiD)	−0.0515***	(0.0006)	−83.23	0.000
Region type (Urban = 1)	−0.0487***	(0.0003)	−179.39	0.000
Age group	+0.0176***	(0.0001)	+159.72	0.000
Caste (numeric)	−0.0000***	(0.0000)	−19.32	0.000
Caste category	+0.0040***	(0.0001)	+52.90	0.000
Religion (numeric)	−0.0061***	(0.0001)	−60.32	0.000
State FE		Yes		
Wave FE		Yes		
Observations		3,874,994		
R^2		0.0677		

Notes: Dependent variable: female LFP (binary). Standard errors clustered at state level. Omitted categories: Andhra Pradesh (state FE), Wave 1 (wave FE). *** $p < 0.01$.

6.7 Individual-Level Determinants

Table 9 examines the role of individual characteristics in shaping female labour force participation.

Table 9: Determinants of Female Labour Force Participation

Variables	Coefficient
Age Group	0.0202***
Caste (Recode)	-0.0217***
State	-0.0022***

*** p<0.01, ** p<0.05, * p<0.1

The results indicate that age is a strong positive predictor of labour market engagement, with older women more likely to participate in economic activity.

But, the social disadvantage remains a significant barrier. Women from historically marginalised caste groups exhibit lower participation rates, highlighting the persistence of structural inequalities in access to employment opportunities.

Taken together, these findings underscore that labour force participation is shaped by a complex interaction of demographic, social and economic factors. Many of these determinants lie outside the direct scope of transport policy, reinforcing the need for complementary interventions to address broader structural constraints.

6.8 Cross-Sectional Determinants of Labour Force Participation

In addition to the causal analysis, we examine cross-sectional correlates of female labour force participation using baseline data. Simple ordinary least squares (OLS) regressions are employed to explore how socio-demographic characteristics are associated with participation outcomes.

The results, summarised in Table 10, reveal several noteworthy patterns. Social identity plays a role: women belonging to Scheduled Caste, Scheduled Tribe or Other Backward Class groups exhibit participation rates that are approximately 2.2 percentage points lower than those of forward caste women.

Education displays a modest negative gradient in the baseline period, with each additional year of schooling associated with a decline of roughly 0.6 percentage points in participation, consistent with the U-shaped relationship documented earlier.

Age is positively associated with participation, with a coefficient of approximately 0.02, indicating that older women are somewhat more likely to engage in economic activity than younger women. This may reflect a combination of economic necessity, life-cycle effects and household dynamics.

These cross-sectional relationships should not be interpreted as causal effects. Rather, they highlight the importance of socio-economic heterogeneity in shaping labour market outcomes and underscore the need to account for these factors when evaluating policy interventions.

Table 10: Cross-sectional determinants of female labour force participation

Variable	Coefficient	Interpretation
SC/ST/OBC	-0.022	Lower participation vs forward caste
Education (years)	-0.006	Negative gradient (U-shape effect)
Age	$+0.02$	Participation increases with age

7 Discussion

The central finding of this paper is that free bus policies have a negative and statistically significant effect on female labour force participation across all specifications. This runs counter to the conventional expectation that reducing transport costs should facilitate women's entry into the labour market. To interpret this result, we propose four potential mechanisms.

First, the **income effect channel**. By eliminating commuting costs, the policy effectively increases women's non-labour income. For some women, particularly those working primarily to cover transport expenses, this may reduce the need to remain employed, allowing them to exit the labour force without a substantial loss in welfare. This interpretation is consistent with the stronger effects observed among urban and highly educated women, who are more likely to be engaged in formal employment with higher commuting costs.

Second, the **mobility reallocation channel**. Improved access to transport may enable women to reallocate their time away from measured formal employment toward unpaid domestic work, informal caregiving, or other informal activities that are not captured in the CMIE definition of labour force participation. The persistence and gradual increase in the estimated effects over time align with the idea of a slow reallocation of labour supply rather than an immediate response.

Third, the **pre-trend caveat**. The statistically significant but relatively small pre-treatment average (-0.0053) calls for some caution in interpretation. Although it is much smaller than the post-treatment average (-0.0238), it suggests that the parallel trends assumption may not hold perfectly. In addition, the positive T_{-1} coefficient could indicate anticipation effects in the period just before the policy was implemented.

Fourth, **data limitations**. The CMIE measure of labour force participation classifies certain forms of home-based work and informal activities as being outside the labour force. If improved mobility allows women to transition from formal employment to home-based entrepreneurship or informal trade, the observed decline in measured labour force participation may overstate any actual decline in welfare. Future research using time-use data would be useful to better understand and decompose these shifts.

Overall, these findings contribute to the literature on transport infrastructure and female labour supply in developing countries (Lei, Desai, and Vanneman, 2019; Chen et al., 2024). By providing large-scale panel evidence from India, the results highlight the importance of income effects and labour reallocation, alongside the more commonly discussed channels of safety and access.

8 Evidence from Previous Evaluations

8.1 Savings and Budget Reallocation

A growing body of empirical and policy-oriented studies highlights that fare-free and subsidised bus schemes generate substantial financial savings for women, particularly those from lower-income households. Table 11 summarises key programme features and reported outcomes across major state-level interventions.

Table 11: Summary of fare-free and subsidised transport schemes for women

Programme	Year	Target Group	Monthly Savings	% Saved	Key Findings
Delhi – Pink Ticket	2019	Women on DTC buses	500– 600	53.5%	Increased ridership; savings used for food, education, health; safety concerns persist
Karnataka – Shakti	2023	Women on state buses	$\geq$ 1,000	50%	Increased work trips; savings redirected; overcrowding and harassment issues
Tamil Nadu (partial)	2020–21	Women travellers	409– 538	30–54%	Partial savings; better service quality; limited coverage
Unsubsidised cities	–	Baseline comparison	0	0	High bus dependence even without subsidies

The evidence suggests that full subsidy schemes deliver the largest direct benefits. Under Delhi's Pink Ticket programme, introduced in 2019, women save approximately 500– 600 per month, amounting to over half of their transport expenditure. These savings are frequently reallocated toward essential household needs, including food consumption, children's education and healthcare. The scheme has also led to a marked increase in women's share of bus ridership, although concerns regarding safety and service quality persist.

Karnataka's Shakti scheme, launched in 2023, exhibits similar patterns but with even higher reported savings, often exceeding 1,000 per month. Survey evidence indicates that a significant proportion of women use these savings to support basic consumption, while also increasing the frequency of work-related travel. At the same time, operational challenges—such as overcrowding and instances of harassment—continue to limit the full realisation of these gains.

Partial subsidy models, such as those implemented in Tamil Nadu and other states, provide more modest financial relief. Women in these settings continue to incur transport expenses, typically in the range of 400– 500 per month, although the relative burden is reduced. These schemes often achieve higher satisfaction with service quality but offer more limited coverage and reach.

Importantly, evidence from unsubsidised cities indicates that reliance on public transport remains high even in the absence of fare reductions. A substantial majority of women continue to depend on buses for daily mobility, suggesting that transport demand is driven not only by affordability but also by structural factors such as accessibility and network coverage.

Taken together, these findings demonstrate that fare-free transport policies deliver meaningful welfare gains through cost savings and improved mobility. However, they also underline that financial relief alone may not be sufficient to generate corresponding improvements in labour market outcomes.

8.2 Mobility Gains and Ridership

Fare-free transport programmes have led to a clear increase in women's use of public transport and an expansion in their mobility. In Delhi, for instance, women's share of daily bus ridership rose from approximately 33 percent prior to the introduction of the Pink Ticket scheme to around 42 percent in 2022–23. Survey evidence further indicates that 23 percent of women increased their frequency of bus use, while 15 percent began using buses only after the policy was implemented.

Similar patterns are observed in Karnataka, where 62 percent of respondents report undertaking additional work-related trips following the introduction of the Shakti scheme. Figure **??** illustrates these shifts in ridership patterns, highlighting the role of fare reductions in expanding women's travel behaviour.

However, increased ridership has not been accompanied by commensurate improvements in service quality. Persistent issues such as overcrowding, unreliable schedules and experiences of harassment continue to constrain the effective use of public transport, particularly during peak hours. As a result, the financial benefits of fare-free travel are often offset by non-monetary costs, including longer travel times, physical discomfort and heightened safety concerns.

8.3 Employment Outcomes and Heterogeneity

The central question remains whether improvements in mobility translate into meaningful gains in women's labour force participation. The available evidence presents a nuanced picture.

An evaluation based on time-use data from Delhi finds that, while women increased the time devoted to paid work by approximately 55 minutes per day following the introduction of the Pink Ticket scheme, the overall probability of labour force participation did not change significantly. This suggests that mobility gains may intensify engagement among existing workers rather than draw new entrants into the labour force.

At the same time, the effects are highly heterogeneous across socio-economic groups. Low-income women without access to private transport experience substantial gains: participation rates increase by 24–39 percentage points, and time spent in paid work rises by more than two hours per day. In contrast, women from higher-income households exhibit little or no change in employment outcomes.

Cross-city comparisons further underscore the importance of local conditions. In Bengaluru, where public transport services are relatively reliable, 21 percent of women report an improvement in their employment situation. In Mumbai, where operational challenges are more pronounced, only 3 percent report similar gains. Figure **??** summarises these differences.

Importantly, even in settings where mobility improves, structural labour market constraints persist. Occupations such as clerical and sales roles remain heavily male-dominated, limiting access for moderately educated women. As a result, the primary beneficiaries of improved mobility tend to be women engaged in self-employment, informal trade or domestic services.

Recent research also emphasises the role of household dynamics and gender norms. Reductions in commuting costs can alter intra-household time allocation, but not always in ways that increase labour supply. In many cases, particularly among married women, time savings are absorbed into unpaid domestic work rather than market activities. By contrast, higher-skilled women are more likely to reallocate time toward employment or education.

Taken together, the evidence suggests that while fare-free transport policies significantly enhance mobility, their impact on labour force participation is conditional on broader structural, institutional and social factors.

9 Discussion

The evidence presented in this study indicates that fare-free bus schemes generate meaningful economic and social benefits, particularly through reductions in direct transport costs and the reallocation of household resources toward essential consumption. By easing financial constraints, these policies enhance women's mobility, expand their spatial access to opportunities, and contribute to greater autonomy in daily decision-making. Importantly, the benefits are strongly redistributive: low-income women and those without access to private transport experience the largest gains, thereby narrowing gender and income disparities in mobility.

At the same time, the impact on labour force participation remains uneven and, in some cases, limited. The most pronounced employment gains are observed among economically disadvantaged women who were previously constrained by the cost of commuting. For women with moderate levels of education or from higher-income households, however, mobility improvements do not translate into comparable increases in labour supply. Structural barriers—such as the limited availability of female-appropriate jobs, persistent occupational segregation, prevailing gender norms and the burden of unpaid care work—continue to constrain labour market engagement. In this context, fare-free transport primarily alleviates mobility constraints and reduces time poverty rather than directly increasing participation rates.

Service quality emerges as a central mediating factor in shaping outcomes. In settings where public transport systems are frequent, reliable and perceived as safe, the potential for employment gains is greater. Conversely, overcrowding, long waiting times and concerns related to harassment significantly reduce the effective benefits of fare-free travel. Women consistently highlight the need for improvements in last-mile connectivity, better lighting and infrastructure at bus stops, and greater sensitivity among transport staff. Addressing these concerns requires sustained investment in transport capacity, maintenance and safety-oriented design.

Finally, the analysis underscores that transport interventions, while important, cannot substitute for broader labour market reforms. Occupational segregation continues to restrict the range of employment opportunities available to women, particularly those with intermediate levels of education. Expanding access to suitable jobs—especially in clerical, sales and technical occupations—alongside policies that support childcare provision and flexible work arrangements, is critical for translating mobility gains into sustained improvements in female labour force participation.

10 Conclusion

This paper evaluates the causal impact of state-level free bus policies on female labour force participation in India, using 33 waves of CMIE CPHS panel data and the Callaway and Sant'Anna (2021) staggered difference-in-differences estimator. The results are consistently negative across all specifications, subgroups, and robustness checks.

Contrary to the premise that motivated these policies, reducing commuting costs does not appear to draw women into the labour market. Instead, the evidence is consistent with women using the income gains from free transport to exit employment, or reallocating their mobility toward non-market activities. The effects are largest for urban women and those with higher education, and grow over time, suggesting a sustained structural shift rather than a transitory adjustment.

These findings suggest that the effectiveness of transport policies depends critically on the broader economic and institutional context in which they operate. To achieve meaningful and sustained increases in female labour force participation, fare-free transport must be embedded within a wider policy framework that includes the expansion of female-friendly employment opportunities, improvements in the safety and reliability of public services, the provision of affordable childcare and the promotion of more equitable gender norms within households and workplaces.

In this sense, transport policies should be viewed not as a standalone solution, but as one component of a comprehensive strategy aimed at reducing both mobility-related and labour market constraints. Only by addressing these dimensions jointly can India fully unlock the economic potential of its female workforce and move toward more inclusive patterns of growth.

Reducing the cost of mobility represents an important step toward advancing gender-inclusive economic growth. The experience of fare-free bus schemes in Delhi, Karnataka and other Indian states demonstrates that such interventions can generate substantial welfare gains by lowering transport costs, expanding women's travel horizons and enhancing autonomy in everyday decision-making.

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